SHIVAM KAMBOJ

Open Quantum Systems · Quantum Error Correction

Merced, CA · (408) 858-3104 · shivam1kamboj@gmail.com · Portfolio · GitHub: @shivam1kamboj

Ph.D. candidate in Physics, UC Merced (expected Fall 2026). Thesis research on ultrastrong light–matter coupling in the quantum Rabi dimer, from the Hamiltonian through symmetry-resolved spectra to open-system dynamics. Author of `nonmarkov_qec`, an open-source package for simulating surface codes under temporally correlated noise. At LLNL, wrote control and process-tomography software for superconducting qudits.

TECHNICAL SKILLS

- **Theory and error correction:** surface-code simulation and decoding; bosonic and cat-state encodings; temporally correlated ($1/f$) noise models; Lindblad master equations; light–matter coupling; normal-mode and Bogoliubov analysis.
- **Numerical methods:** symmetry-resolved sparse exact diagonalization; Hilbert-space truncation and convergence checks; Monte Carlo threshold estimation with bootstrap confidence intervals; linear-inversion process tomography.
- **Software:** Python, NumPy, SciPy, QuTiP, Stim, PyMatching, Qiskit; SLURM, Linux, Git, GitHub Actions; `pytest`, `mypy`, `ruff`; Flask, Gemini API, Google Cloud; COMSOL RF and AC/DC.

RESEARCH EXPERIENCE

Tian Group, University of California, Merced — *Graduate Student Researcher (advisor: Prof. Lin Tian)* **May 2023 – Present**

- Model ultrastrong light–matter coupling in the quantum Rabi dimer, connecting Hamiltonian structure to spectra, collective modes, and quantum-state properties.
- Combine normal-mode analysis and sparse exact diagonalization to resolve low-energy states by parity and site-exchange symmetry, distinguish true from avoided level crossings, and characterize a candidate cat-state doublet.
- Simulate state evolution, entanglement, and dissipation in QuTiP; check convergence in Hilbert-space truncation and run checkpointed parameter studies on SLURM clusters.
- Build COMSOL electromagnetic models of a proposed superconducting implementation; extract capacitances, mode frequencies, impedances, and inductive participation to assess which interactions are feasible.
- Proposed extension (independent): test whether the doublet's parity structure can support a noise-biased bosonic qubit.

Lawrence Livermore National Laboratory — *Physics Graduate Intern, Quantum Coherent Device Physics Group* **May – Aug 2024**

- Wrote Python software for pulse-level control of superconducting qudits and evaluated gate sequences with quantum process tomography.
- Built a generalized linear-inversion tomography pipeline (pseudo-inverse reconstruction) that turns measured data into process maps.
- Automated Ramsey and T_1/T_2 fits to give quantitative feedback during control experiments.

Sharping Group (UC Merced) and Hurst Group (SJSU / UC Merced) — *Research Collaborator; Graduate Researcher* **Jun 2021 – Present**

- Develop Hamiltonian models of coupled YIG–cavity (magnon–photon) systems with inter-cavity coupling and parametric driving, working with an experimental group.
- Derived dispersion relations and numerical solvers for 2D spin-torque oscillator arrays and studied how edge modes respond to disorder (first-author paper, *Phys. Rev. B*).

SOFTWARE

`nonmarkov_qec` — *open-source Python package for surface-code simulation under correlated noise; released on PyPI* **2026**

- Generates Ornstein–Uhlenbeck and sum-of-OU ($1/f$) dephasing trajectories, injects them into precompiled Stim circuit templates (about 35× faster than rebuilding circuits), decodes with PyMatching, and estimates thresholds by Monte Carlo with parametric-bootstrap 95% confidence intervals.
- Compared correlated dephasing with a statistically matched white-noise baseline on rotated surface codes ($d = 3, 5$) to separate correlation effects from changes in error strength. In the regime studied, correlations did not lower the threshold, consistent with Kam et al. (*Quantum Sci. Technol.*, 2025).
- 74 unit and analytic-validation tests; strict type checking, linting, and CI with automated releases.

Physics Concept Analyzer Tool (PCAT) — *Python, Flask, Gemini API, Google Cloud* **2025**

- Built and deployed a web application on Google App Engine that guides physics students through concepts, assumptions, and model construction before calculation; piloted in UC Merced physics courses.

EDUCATION, PUBLICATIONS AND AWARDS

Ph.D. in Physics, UC Merced (expected Fall 2026) · **M.S. in Physics**, San Jose State University (2022)

1. S. Kamboj, R. A. Duine, B. Flebus, and H. M. Hurst, "Oscillatory edge modes in two-dimensional spin-torque oscillator arrays," *Phys. Rev. B* **109**, 094436 (2024).

2. S. Kamboj and L. Tian, quantum Rabi dimer and candidate bosonic encodings (in preparation).

3. S. Kamboj, J. Beurkens, J. Sharping, and L. Tian, "Superconducting realization of the quantum Rabi dimer" (in preparation).

Awards: Outstanding Graduate Teaching Award, UC Merced (2026); Outstanding Teaching Assistant Award, SJSU (2022); Physics Graduate Group Summer Fellowship (2025, 2026).